

Modeling the Effect of Pitch Sequencing and Batter-Specific Adjustments on Swing-and-Miss Outcomes in MLB

Group 10: Harsh Dave, Nick Griffin, Jack Leidlein, Hector Carrillo

Spring 2026

1 Introduction

The rise of advanced tracking systems in Major League Baseball (MLB) has fundamentally transformed how teams evaluate pitching performance. Since the introduction of Statcast in 2015, every pitch thrown in an MLB game is measured with unprecedented precision—release speed, spin rate, movement profile, release point, and extension are all captured at the sub-millisecond level (Major League Baseball, 2024). This wealth of data has fueled a revolution in pitcher development, with front offices investing heavily in pitch design laboratories and data-driven coaching programs aimed at maximizing the effectiveness of each pitch in a pitcher’s arsenal. A central question in modern baseball analytics is: *what attributes make a pitch difficult to hit?* Intuitively, pitches with higher velocity, greater movement, and longer extension should be more effective. However, standard logistic regression models that predict swing-and-miss probability based solely on these physical characteristics often yield inconclusive results. In such models, the probability of a whiff, $p_i = P(Y_i = 1)$, is typically modeled as:

$$\log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 \text{Speed}_i + \beta_2 \text{IVB}_i + \beta_3 \text{HB}_i + \beta_4 \text{Ext}_i + \beta_5 \text{Arm}_i + \epsilon_i \quad (1)$$

where **IVB** denotes induced vertical break, **HB** horizontal break, **Ext** release extension, and **Arm** arm angle. Previous studies have found that main effects and simple interactions in this framework are often statistically insignificant or fail to explain a substantial portion of the variance in outcomes. The limitation of this approach is that it treats each pitch as an independent, context-free event, ignoring two critical sources of variation: **sequencing** and **batter heterogeneity**. First, pitches are part of a strategic sequence. A pitch’s effectiveness is often determined not by its raw metrics, but by how well it tunnels with the preceding pitch. For example, a breaking ball may induce more whiffs when it follows a fastball due to the disruption in batter timing and visual processing—a phenomenon known as “pitch tunneling.” Second, batters differ significantly in their baseline contact rates. Elite contact hitters rarely swing and miss regardless of pitch quality, while

high-strikeout batters may whiff at average pitches. Neglecting these batter-specific tendencies introduces substantial noise into the analysis, obscuring the true relationship between pitch quality and outcomes. To address these limitations, we propose a refined modeling framework that explicitly incorporates pitch sequencing and batter-specific adjustments. Our research integrates lagged pitch information to capture sequencing effects and employs hierarchical modeling to account for batter-level variability. Specifically, our research question is:

How do pitch sequencing (the type, location, and result of the previous pitch) and batter-specific tendencies jointly influence the probability of a swing-and-miss, beyond the physical characteristics of the pitch itself?

To investigate this, we propose the following objectives:

- 1. Quantify Sequencing Effects:** Investigate whether the previous pitch type and result significantly affect the whiff probability of the current pitch. We hypothesize that specific sequence combinations (e.g., Fastball → Slider) will show significant interaction effects.
- 2. Model Batter Heterogeneity:** Implement a Generalized Linear Mixed Model (GLMM) where individual batters are treated as random effects. This approach separates the variance attributable to the pitch from the variance attributable to the batter’s skill.
- 3. Compare Predictive Performance:** Evaluate whether the Sequencing and Mixed-Effects models provide superior fit and predictive accuracy compared to the baseline pitch-characteristics-only model.

2 Research Strategy

2.1 Data

2.2 Planned Analysis

References

- Gelman, A. and J. Hill (2006). *Data analysis using regression and multilevel/hierarchical models*. Cambridge university press.
- Major League Baseball (2024). Statcast search. https://baseballsavant.mlb.com/statcast_search. Pitch-by-pitch tracking data for all MLB games.
- McCullagh, P. and J. A. Nelder (2019). *Generalized linear models*. Routledge.
- Peta, J. and Contributors (2024). pybaseball: Pull baseball data in python. <https://github.com/jldbc/pybaseball>. Python package for accessing Statcast, FanGraphs, and Baseball Reference data.